Step 10 Final Robustness Report

This report implements the final participant-fixed-effects robustness check using the same Step 9 duration-aware front ROI outcome and the same included cohort, with 36 participant-sessions, 30 questions, and 1075 trials. The primary inferential specification used Two-way cluster (participant + question).

Primary Model

Metric	Value
Robust SE type	Two-way cluster (participant + question)
Included participants	36
Included questions	30
Included trials	1075
Condition beta	3.016e-08
Condition SE	2.351e-08
Condition test statistic	1.283
Condition p-value	0.209749
Condition 95% CI	[-1.793e-08, 7.824e-08]
Global block-order beta	4.422e-09
Within-block-position beta	-6.034e-09
Word-count beta	-4.611e-08
ENEM-correctness beta	3.251e-09

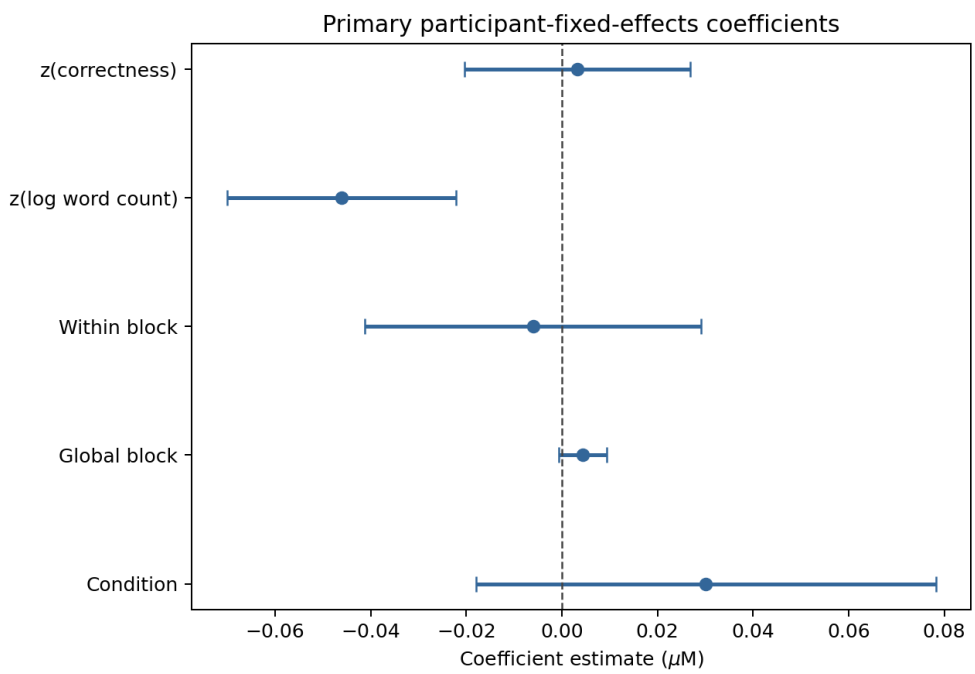

Figure 1: Coefficient plot for the primary participant-fixed-effects model.

Sensitivity Models

Model	SE type	Beta / range	SE	p	95 Primary participant-fixed-effects model
Two-way cluster (participant + question)	3.016e-08	2.351e-08	0.209749	[-1.793e-08, 7.824e-08]	
Participant-clustered participant-fixed-effects model	Participant-clustered	3.016e-08	2.058e-08	0.151850	[-1.163e-08, 7.194e-08]
Question-clustered participant-fixed-effects model	Question-clustered	3.016e-08	2.449e-08	0.228028	[-1.993e-08, 8.024e-08]
HC3 participant-fixed-effects model	HC3 heteroskedasticity-robust	3.016e-08	2.212e-08	0.173034	[-1.324e-08, 7.356e-08]
Fixed-window front ROI benchmark under participant fixed effects	Two-way cluster (participant + question)	2.235e-08	1.780e-08	0.219180	[-1.405e-08, 5.874e-08]
No-override participant-fixed-effects model	Two-way cluster (participant + question)	1.910e-08	2.086e-08	0.367415	[-2.357e-08, 6.178e-08]
Back ROI benchmark under participant fixed effects	Two-way cluster (participant + question)	1.388e-08	1.119e-08	0.224774	[-9.009e-09, 3.678e-08]
Leave-one-question-out range	Primary robust SE	1.885e-08 to 4.181e-08; p range 0.079410 to 0.412718			
Leave-one-participant-out range	Primary robust SE	2.196e-08 to 4.119e-08; p range 0.067180 to 0.292688			

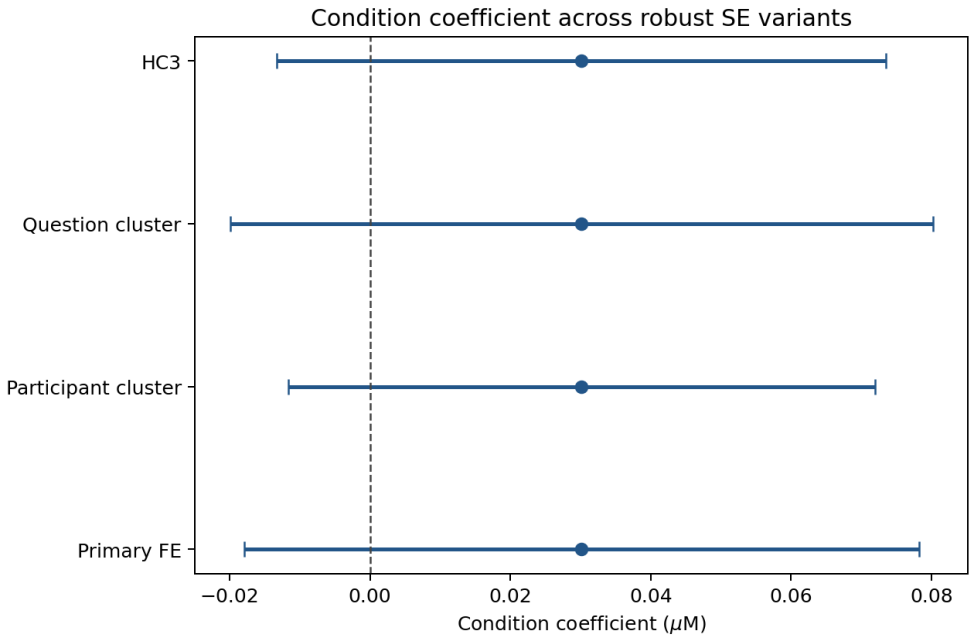

Figure 2: Condition coefficient under the robust standard-error variants.

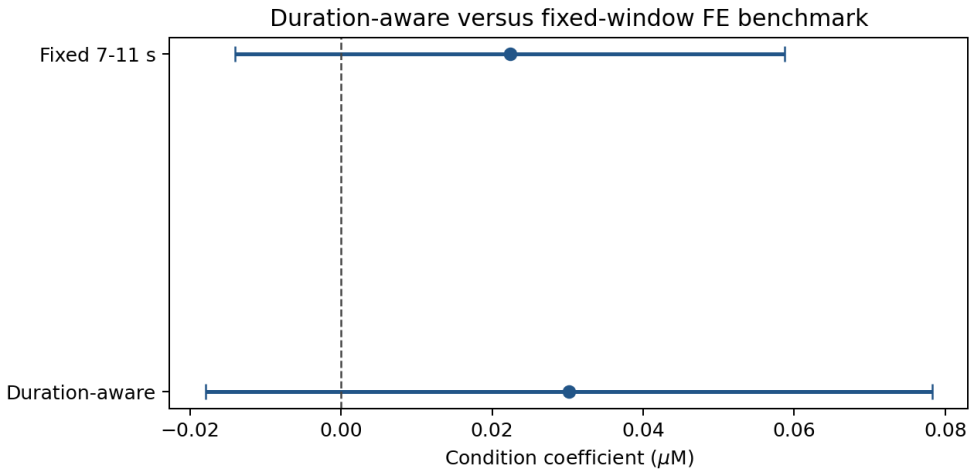

Figure 3: Final fixed-window versus duration-aware benchmark under participant fixed effects.

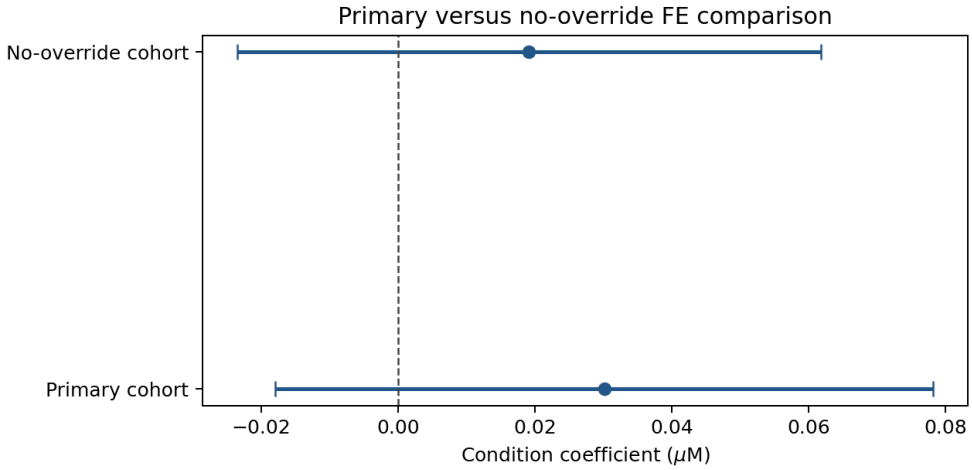

Figure 4: Primary cohort versus no-override sensitivity comparison.

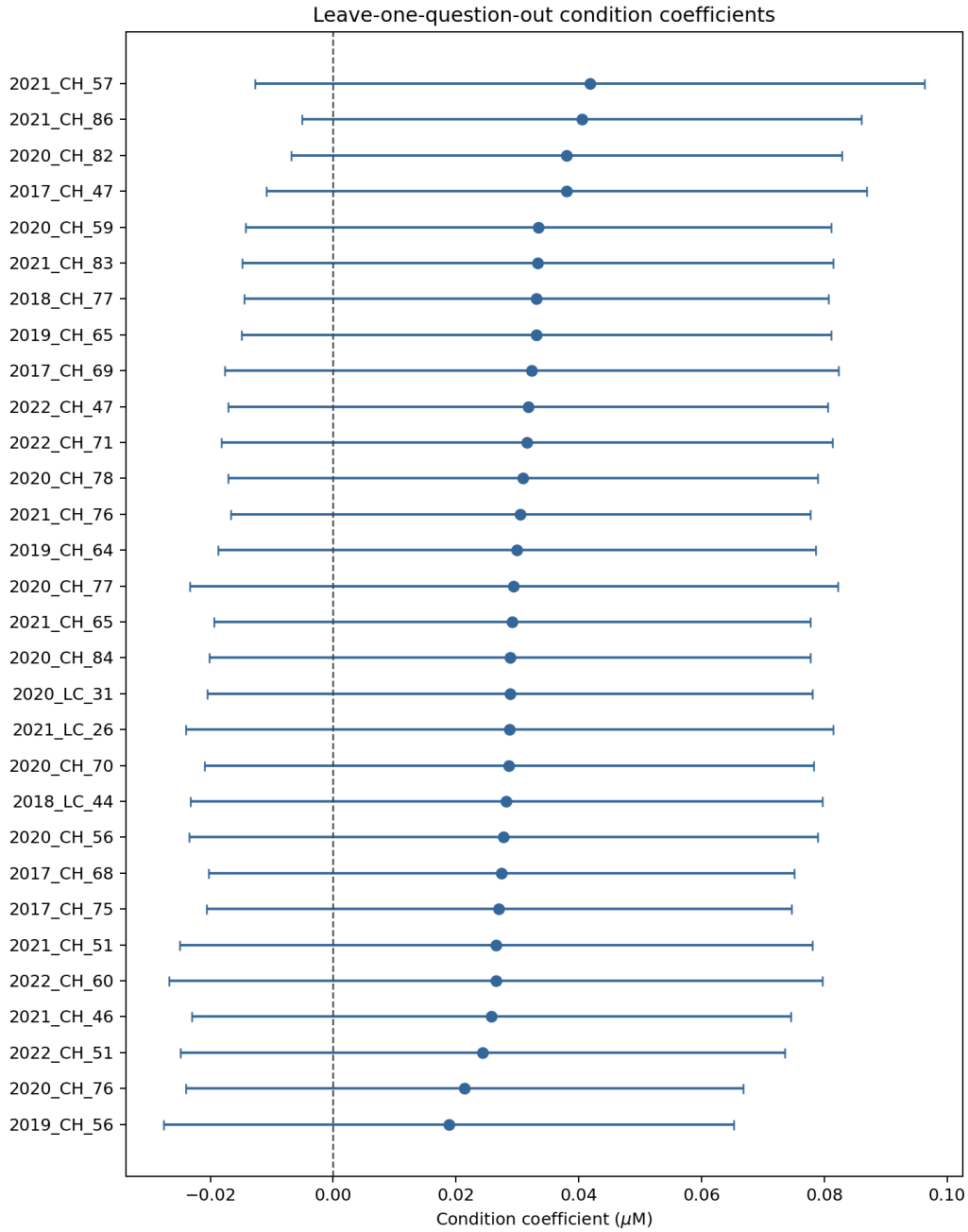

Figure 5: Leave-one-question-out condition coefficients with 95% confidence intervals.

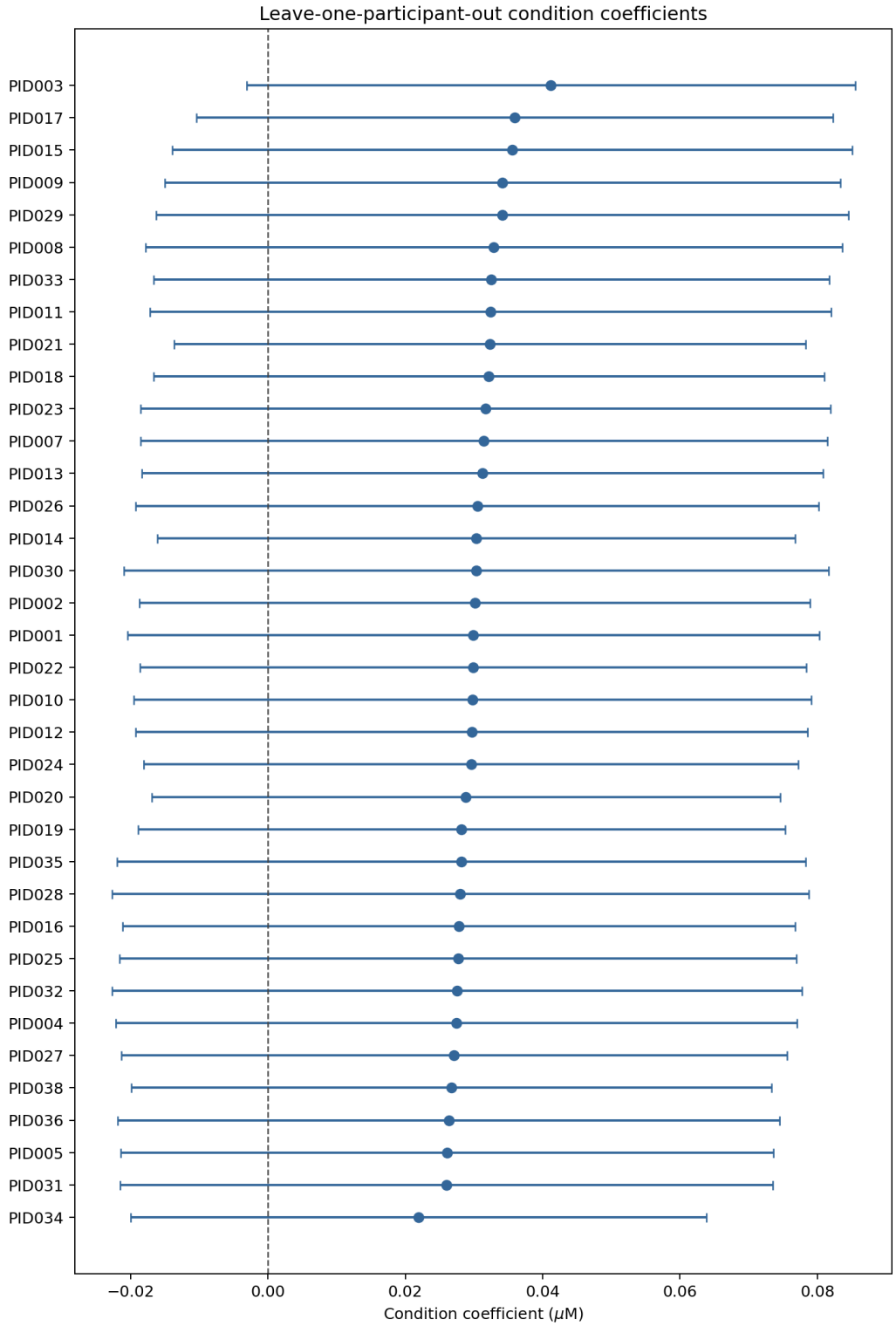

Figure 6: Leave-one-participant-out condition coefficients with 95% confidence intervals.

Warnings And Notes

- No Step 10-specific warnings were generated.

Final Conclusion

In the final participant-fixed-effects robustness analysis, the abstract-versus-concrete effect in the left frontal ROI was not statistically significant (beta=3.016e-08, SE=2.351e-08, p=0.209749, 95